

Vehicle Classification Based on Feature Selection With Anisotropic Magnetoresistive Sensor

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Abstract—Vehicle classification based on magnetic sensors can be effectively applied to intelligent transportation and realize intelligent management of traffic. The use of representative vehicle signal features is a prerequisite and guarantee for accurate vehicle classification. This paper uses a single 3-axis magnetic sensor to acquire vehicle signals and extract a large number of features from the vehicle signals. In order to obtain a set of features that are simple and without loss of classification accuracy, we propose a Filtering algorithm based on Feature Pairing Elimination (FPE-Filter). In addition, choosing the right classifier is also crucial for models with high classification accuracy. In this paper, we compare four common classification models: SVM, RF, KNN, and C4.5. The experimental results show that the SVM performs best and the classification accuracy reaches 95%.

Index Terms—Anisotropic magnetoresistive sensor (AMR), vehicle classification, feature selection, ReliefF.

I. INTRODUCTION

WITH the rising of the people's travel demand, intelligent transportation technology plays an increasingly important role in the optimal management of traffic. The detection of vehicles and the classification of vehicles are an indispensable part of intelligent transportation technology. For vehicle detection, inductive loop detectors and cameras are commonly used for detection. However, it is well known that the installation of the inductive loop detector has a high cost, and the flaw of the camera being vulnerable to the climatic environment is still its biggest shortcoming. Compared with induction coils and cameras, the magnetic sensor's small size, easy installation, low cost and strong climate adaptability make it more and more widely used. Many methods have studied the detection of vehicles based on magnetic sensors, such as [1]–[3]. The focus of this paper is on the vehicle classification. Therefore, after comparing various vehicle detection algorithms, the double window detection algorithm proposed in [3] is used for vehicle detection and signal acquisition because of its strong stability and higher accuracy.

In many studies, the number of sensors used by vehicle detection nodes is one or two, both of which have advantages

and disadvantages. The use of two magnetic sensors provides an important feature of vehicle length, while the use of a single magnetic sensor reduces costs. In addition, the vehicle signal features extracted by different literature are also different. For example, the work detailed in [4] uses two 3-axis AMR sensors to collect the magnetic field disturbance signal of the vehicle. The vehicle length, average energy and the peak number of the hill pattern are taken as features of the vehicle. The process in [5] uses a single 3-axis AMR sensor to extract the duration of the signal, the signal energy, the average energy of the signal, the positive-negative energy ratio of the X-axis signal, and the positive-negative energy ratio of the Y-axis signal as features of the vehicle. Other similar methods such as [6]–[8] extract the average perpendicular magnetic height, differential energy, variance, maximum value, number of extreme values, and other features of the signal.

On the other hand, most methods directly apply the obtained feature set to the classification algorithm after feature extraction. However, not all of these features are equally important for a specific task. Some of them may be redundant or even irrelevant [9]. These redundant features can not only impair the accuracy of the classification algorithm, but also affect the execution efficiency of the algorithm. One way to solve this problem is feature selection. Feature selection is the process of selecting a smaller features subset that is sufficient to accurately predict the class labels of the training set [10].

This paper focuses on the effectiveness of feature selection for vehicle classification based on 3-axis magnetic sensors. First, the vehicle is detected using double window detection algorithm and the original vehicle signal is acquired. Second, different features for characterizing the original vehicle signal are extracted to obtain a feature set for vehicle classification algorithm training and testing. Moreover, the established feature selection algorithm selects the optimal feature subset without redundant features in the feature set. Finally, the classification algorithm is used to train the obtained feature subsets to obtain the optimal classification model.

The remainder of this paper is organized as follows. Section II introduces the foundation of vehicle classification using magnetic sensor. Section III introduces the feature extraction of vehicle signals. Section IV describes the principles of the novel feature selection algorithm. Section V introduces how to build the classification models. The experimental results of vehicle classification are given in Section VI. Section VII gives the conclusions of this paper.

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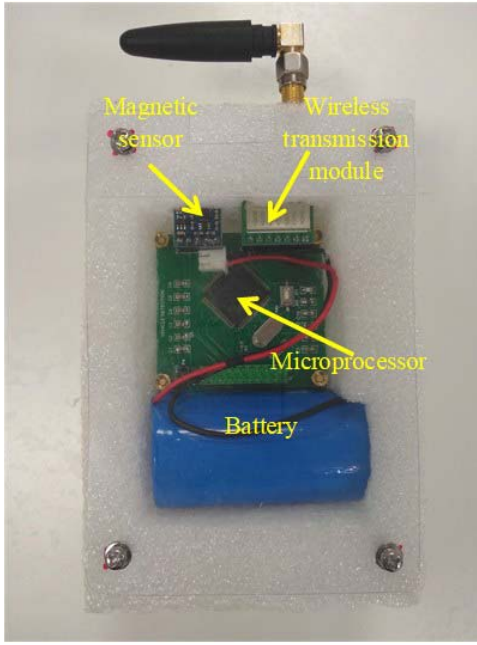


Fig. 1. Sensor node.

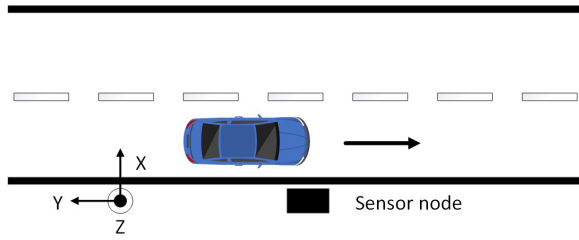


Fig. 2. Sensor node configuration diagram.

II. FOUNDATION OF VEHICLE CLASSIFICATION USING AMR SENSOR

The earth provides a uniform magnetic field over a wide area. A car can cause magnetic field disturbances in specific areas due to its own metal material. Under the same measurement conditions, the magnetic field disturbance caused by different types of vehicles will be different. Therefore, the magnetic field disturbance information of different vehicles collected using the magnetic sensor can be used for vehicle classification.

Fig.1 shows a sensor node for vehicle detection and signal collection. The node contains an STM32F107ZET6 microprocessor, a 3-axis magnetic sensor module and a wireless transmission module. Among them, the 3-axis magnetic sensor we used is Honeywell triaxial HMC5883L. This sensor can achieve 5 milligauss resolution in ± 8 gauss field. In this paper, the sample rate of the magnetic sensor is 75 Sa/s.

Fig.2 shows the layout of the sensor nodes. The X-axis is perpendicular to the direction of the lane. The Y-axis is contrary to the direction of the lane and the Z-axis is perpendicular to the road surface and upward.

TABLE I
FEATURES EXTRACTED FROM VEHICLE SIGNALS

	Feature	Symbol
1	Maximum	$X_{max}, Y_{max}, Z_{max}$
2	Range	X_R, Y_R, Z_R
3	Relative position of maximum	$X_{P_{max}}, Y_{P_{max}}, Z_{P_{max}}$
4	Relative position of minimum	$X_{P_{min}}, Y_{P_{min}}, Z_{P_{min}}$
5	Mean	$X_{mean}, Y_{mean}, Z_{mean}$
6	Ratio of positive and negative energy	$X_{P_n}, Y_{P_n}, Z_{P_n}$
7	Number of local maxima	$X_{N_{max}}, Y_{N_{max}}, Z_{N_{max}}$
8	Number of local minima	$X_{N_{min}}, Y_{N_{min}}, Z_{N_{min}}$
9	Variance	$X_{var}, Y_{var}, Z_{var}$
10	Approximate entropy	$X_{ApEn}, Y_{ApEn}, Z_{ApEn}$
11	Crossing mean counts	$X_{cross}, Y_{cross}, Z_{cross}$
12	Average energy	Ev

III. FEATURE EXTRACTION

Generally, a huge number of data points are included in the vehicle signal. Redundant data slow down the classification process and often cause inaccurate results. To reduce the data dimensionality and for a better performance, feature extraction techniques are often used in this stage [11]. In this paper, the features of the signals acquired in the three directions of the 3-axis magnetic sensor will be extracted. The obtained features will be combined into a feature set to describe the vehicle type information for a car. It should be noted that the signal described in this paper is the signal after the background magnetic field is removed. The features used in this paper are shown in TABLE I. Among these features, maximum, range, relative position of maximum, relative position of minimum, number of local maxima, and the number of local minima can describe the difference in local structure between different signals, while the mean, ratio of positive and negative energy, variance, approximate entropy, crossing mean counts, and the average energy can reflect the inherent differences in the overall structure between different signals. After extracting each set of features for each vehicle, we include each set of features in a feature set. This feature set will be used as the operational object of the subsequent feature selection algorithm.

In TABLE I, X_{max} , Y_{max} , and Z_{max} represent the maximum values of the signals collected in the X-axis, Y-axis, and Z-axis

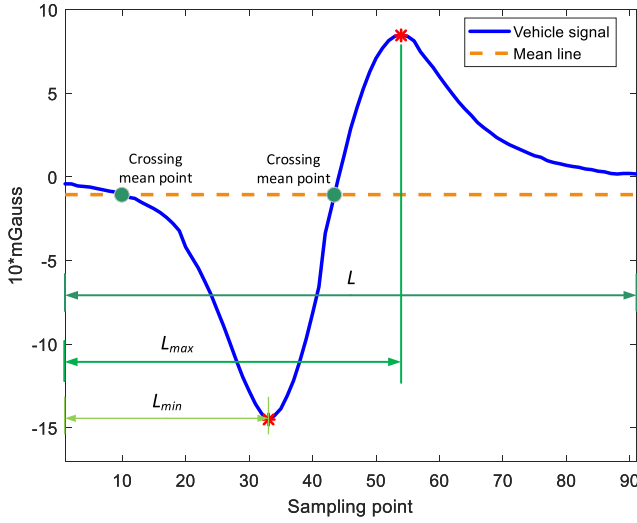


Fig. 3. Interpretation of features.

directions of the magnetic sensor, respectively. The same principle is also applied to other feature symbol. Specifically, the relative position of maximum and the relative position of minimum are defined as

$$\begin{cases} X_{P_{max}} = \frac{X_{L_{max}}}{X_L} \\ Y_{P_{max}} = \frac{Y_{L_{max}}}{Y_L} \\ Z_{P_{max}} = \frac{Z_{L_{max}}}{Z_L} \end{cases} \quad (1)$$

where, L_{max} and L_{min} indicate the distances of the maximum point and the minimum point to the detection starting point, L represent the length of the signal (Fig.3).

The ratio of positive and negative energy and the average energy are defined as

$$\begin{cases} X_{P_n} = \frac{\sum_{X(k)>0} X(k)}{\sum_{X(k)<0} X(k)} \\ Y_{P_n} = \frac{\sum_{Y(k)>0} Y(k)}{\sum_{Y(k)<0} Y(k)} \\ Z_{P_n} = \frac{\sum_{Z(k)>0} Z(k)}{\sum_{Z(k)<0} Z(k)} \\ Ev = \frac{\sum_{k=1}^L (X(k)^2 + Y(k)^2 + Z(k)^2)}{L} \end{cases} \quad (2)$$

where, $X(k)$, $Y(k)$, and $Z(k)$ indicates the k -th value of the signal in the three directions of the magnetic sensor.

Crossing mean counts feature is defined as the times of vehicle signal crossing the mean line [7], which is two in the Fig.3. Approximate entropy is an indicator to describe the complexity of a signal. The specific calculation is described in [12].

In conclusion, we extracted a total of 34 features from the sensor's triaxial signal:

$$F = [X_{max}, X_R, X_{P_{max}}, X_{mean}, X_{P_n}, X_{N_{max}}, X_{N_{min}}, X_{var}, X_{ApEn}, X_{cross}, Y_{max}, Y_R, Y_{P_{max}}, Y_{mean}, Y_{P_n}, Y_{N_{max}}, Y_{N_{min}}, Y_{var}, Y_{ApEn}, Y_{cross}, Z_{max}, Z_R, Z_{P_{max}}, Z_{mean}, Z_{P_n}, Z_{N_{max}}, Z_{N_{min}}, Z_{var}, Z_{ApEn}, Z_{cross}, Ev] \quad (3)$$

IV. FEATURE SELECTION

After the process of feature extraction, the signal samples for each vehicle will be described by 34 features. Designing a classifier with a large number of features under a limited number of samples can make the computational overhead too large and affect classification performance. Therefore, it is particularly important to select a feature subset that is smaller and more stable than the entire feature set by feature selection. Feature selection can be divided into three general categories: 1) filters 2) wrappers 3) embedded techniques. Filter methods represent the simplest and fastest approaches. They sort and selects features based on statistical measures (e.g., correlation, entropy). The main disadvantage of the filter approach is that it totally ignores the effects of the selected feature subset on the performance of the induction algorithm [13]. On the other hand, wrapper uses a search algorithm to search through the space of possible features and evaluate each subset by running the follow-up learning model on the feature subset. Wrapper can be computationally expensive and have a risk of overfitting to the model [14]. Finally, the main idea of embedded selection is to integrate the feature selection process with the learning process of the selected classifier. The CART is a representative example of this method. Embedded techniques are more computationally efficient than wrappers as they only need one model to produce results, but because of that, they are limited to the biases and weaknesses of a specific classifier [15]. Considering the advantages and disadvantages of the three feature selection methods, we propose a filtering selection method based on Feature Pairing Elimination – FPE-Filter. This method not only has the advantages of the filtering method, but also removes the redundant features in the feature set, avoiding the high computational cost of the wrapper algorithm.

FPE-Filter is divided into two steps: First, the filtering algorithm is used to perform preliminary filtering on the feature set, and secondly, the redundant features in the filtered feature set are deleted.

A. Preliminary Filter

In this paper, ReliefF [16] algorithm is used to filter the feature set. Although the ReliefF algorithm is simple, its high

efficiency and good running results make it widely used. The feature selection criteria of the ReliefF algorithm is based on the ability of features to distinguish between close-range samples. When dealing with multi-classification problems, ReliefF randomly selects an instance R , and then searches for k of its nearest neighbors from the same class, called nearest hits H_j , and also k nearest neighbors from each of the different classes, called nearest misses $M_j(C)$ [17]. The weight calculation formula of feature A is as follows

$$W(A)^{i+1} = W(A)^i - \sum_{j=1}^k \frac{\text{diff}(A, R, H_j)}{m \cdot k} + \sum_{C \neq \text{class}(R)} \frac{\frac{p(C)}{1-p(\text{class}(R))} \cdot \sum_{j=1}^k \text{diff}(A, R, M_j(C))}{m \cdot k} \quad (4)$$

where, m represent the number of samples, $\text{class}(R)$ is the class label of the instance R , $p(C)$ is the probability of class C . $\text{diff}(A, R_1, R_2)$ indicates the difference between the samples R_1 and R_2 on the feature A . The calculation formula is as follows

$$\text{diff}(A, R_1, R_2) = \frac{|R_1(A) - R_2(A)|}{\max(A) - \min(A)} \quad (5)$$

Therefore, the preliminary filtering based on the ReliefF algorithm is given below:

- 1) The weight of each feature in the feature set F is obtained by the ReliefF algorithm.
- 2) The features in the feature set F are sorted in descending order by reliefF weights.
- 3) The weight threshold W_t needs to be set, and if the feature's reliefF weight is lower than W_t , the feature will be deleted from the feature set F . Finally, we can get a new feature set F' .

The ReliefF filtered feature subset F' is not optimal and may contain redundant features. Therefore, further screening and filtering is required for the obtained feature subset.

B. Remove Redundant Features

Redundant features can be found by calculating the correlation between features. This paper uses the Pearson correlation coefficient to measure the correlation between two features. The calculated value of the Pearson correlation coefficient falls within the interval $[-1, 1]$. When the correlation coefficient is 1, it becomes completely positive correlation; when the correlation coefficient is -1 , it becomes completely negative correlation; the larger the absolute value of the correlation coefficient is, the stronger the correlation is; the closer the correlation coefficient is to 0, the weaker the correlation. The formula for calculating the Pearson coefficient between two features is as follows

$$r(A_1, A_2) = \frac{\text{Cov}(A_1, A_2)}{\sqrt{\text{Var}(A_1)} \cdot \sqrt{\text{Var}(A_2)}} \quad (6)$$

where, $\text{Cov}(X, Y)$ is the covariance between X and Y , $\text{Var}(X)$ is the variance of X .

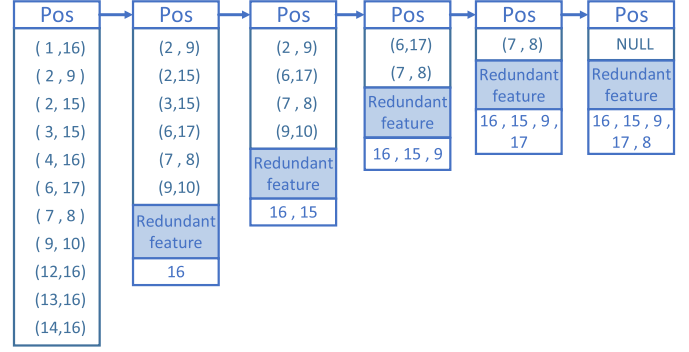


Fig. 4. Schematic diagram of selecting redundant features.

The correlation coefficient threshold R_t needs to be set before the filtering of the redundant features. In this paper, if the absolute value of the correlation coefficient between the two features is greater than the threshold R_t , the two features are said to be *paired*. In addition, the concept of Feature Pairing Number (FPN) is introduced on the basis of feature pairing. If two different features are *paired*, the FPN of both features is increased by one. The initial FPN of each feature is zero. Obviously, if the FPN of a feature is larger, the correlation between it and other features is greater, and the redundancy of the feature is stronger. The idea of the feature selection algorithm proposed in this paper is mainly to count the FPN of each feature, and to simplify the feature set by removing the features corresponding to the larger FPN.

Figure 4 is taken as an example to show the selection process of redundant features. The feature pair (m, n) is used to represent the m -th feature and the n -th feature pair in the feature subset F' . Because the features have been sorted in descending order by weight in the reliefF filtering stage, the larger the feature number, the smaller the corresponding reliefF weight. In order to facilitate the calculation of FPN, we store all *paired* feature pairs in an array called *Pos*. The feature with the largest FPN will be selected and the feature pair containing the feature will be removed from the *Pos*, and meanwhile the feature will be removed from the feature subset F' . The determination of the maximum FPN and the deletion of the feature corresponding to the maximum FPN are repeatedly performed until *Pos* is empty. In a certain selection, if multiple features obtain the largest FPN value, the feature with the lowest reliefF weight among the features will be selected as redundant feature and deleted in the feature set. The pseudo code of the specific algorithm is shown in Algorithm 1.

The algorithm finally gets an optimal subset F_g that without redundant features. In conclusion, this feature selection algorithm requires two input parameters, one is the reliefF weight threshold W_t , and the other is the Pearson correlation coefficient threshold R_t .

V. BUILDING CLASSIFICATION MODELS

The classification results obtained by different classifiers under the same data set are also different. So in order to find the classifier that is most suitable for vehicle classification, we

Algorithm 1 Basic Idea for Removing Redundant Features**Input:** Feature subset F' filtered by ReliefF**Output:** Optimal subset F_g without redundant features

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1 Initialize  $P = \{P_n = 0, n = 1, 2, \dots, N\}$ ,  $Pos \leftarrow \emptyset$ .  $P_n$ 
  is the FPN of the  $n$ -th feature, and  $N$  is the number of
  features in  $F'$ ;
2 The Pearson correlation coefficient  $r_{ij}$  between two
  different features in  $F'$  is calculated;
3 if  $abs(r_{ij}) \geq R_t$  then
4    $Pos \leftarrow (i, j)$ ;
5    $P_i = P_i + 1, P_j = P_j + 1$ ;
6 while  $Pos \neq \emptyset$  do
7   Find the maximum value of  $P$ ;
8   if (The number of maximum values in  $P$ )  $\neq 1$  then
9     The ReliefF weights of all the features that obtain
     the FPN maximum value are compared, and the
     feature with the smallest ReliefF weight is taken as
     the redundant feature;
10  else
11    The feature corresponding to the maximum value
    is taken as the redundant feature;
12  Delete the feature pairs in the  $Pos$  that contains the
    redundant feature;
13  Delete the redundant feature in  $F'$ ;
14  Update the FPN according to the  $Pos$ ;
15 return  $F'$ ;

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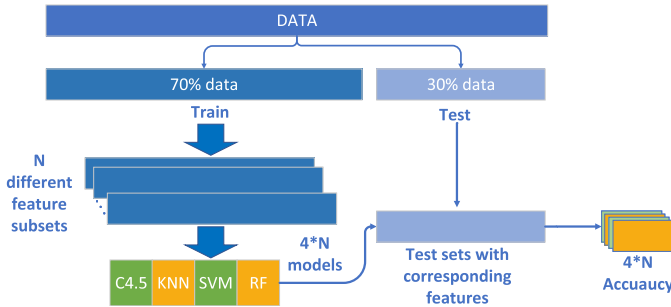


Fig. 5. Schematic diagram of constructing classification models.

compare four commonly used classifier models: C4.5, KNN, SVM, and Random Forest (RF).

In this paper, the vehicle types are divided into three categories according to the size and structure of the vehicle: TRUCK, Saloon and SUV (SAS), and BUS. We collected a total of 2436 vehicle signals (BUS-738, SAUV-987, TRUCK-711) and extracted 34 signal features for each vehicle signal. In order to get a reliable classification result, the data set is divided into two parts. As shown in Figure 5, 70% of the data is used to train the model and the remaining 30% is used as an independent test set. Then, we apply FPE-Filter to the train set. Under the condition of $W_t \in [0.04, 0.1]$ (step: 0.01), $R_t \in [0.55, 0.85]$ (step: 0.1), 28 sets of feature subsets are obtained (Some of these feature subsets are likely to be the same). Next, we constructed different classification

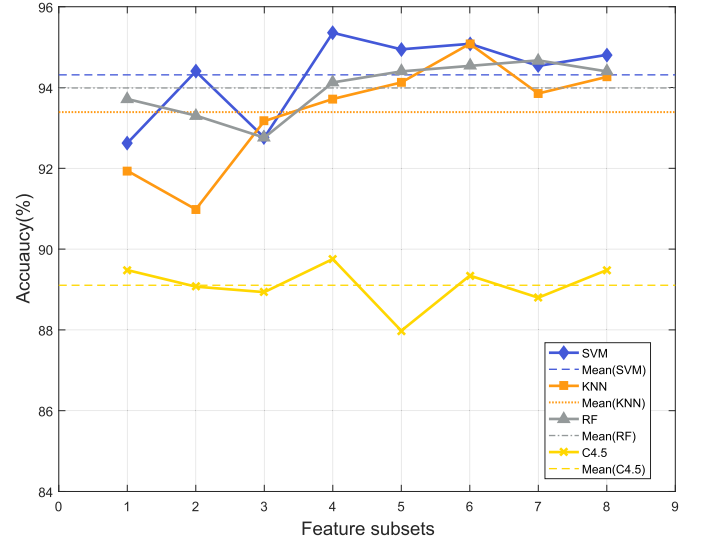


Fig. 6. Accuracy of feature subsets.

models with different classifiers on these feature subsets. In the process of model building, grid search is used to search for optimal model parameters, and 5-fold cross-validation is used to evaluate the model classification accuracy. Finally, the test set was used to verify the model performance.

VI. EXPERIMENTAL RESULTS AND ANALYSIS

This paper uses the classification accuracy ACC as the indicator to evaluate the classification model. It is defined as

$$ACC = N_{correct} / N_{total} \quad (7)$$

where, $N_{correct}$ is the number of correctly classified cars, and N_{total} is the total number of tested vehicles.

A total of 8 different feature subsets were generated in this experiment, as shown in Table II. Fig. 6 shows the classification accuracy and average classification accuracy of C4.5, KNN, SVM, and RF under different subsets. From the perspective of the classification average, the SVM performs best in four classifiers, followed by RF, and C4.5 is the worst performer. In addition, the optimal feature subsets corresponding to different classifiers are also different. For example, SVM achieves the highest classification accuracy in the fourth feature subset, while KNN obtains the optimal value in the sixth feature subset. Moreover, We also compared the results of the classifiers between the FPE-Filter based and the no-feature-selection under the conditions of feature extraction and no feature extraction, respectively, as shown in Table III. The construction method for the data set under the no feature extraction condition is as follows.

Since the information of each vehicle is characterized by the magnetic field disturbance signal in the three directions of X, Y, Z, we serially connect the signals into a new signal S in the order of X, Y, and Z, that is, $S_i = [X_i, Y_i, Z_i]$, where i represents the data sample for the i -th car. Considering the length of each S_i is different, we consider the length of the longest signal as the length reference, and the other shorter signals are padded with the zero-padding method to keep the

TABLE II
FEATURE SUBSETS SELECTED BY FPE-FILTER

Feature sets	Features	X_{max}	$X_{P_{max}}$	$X_{P_{min}}$	$Z_{P_{min}}$	$Y_{P_{min}}$	Y_{max}	$Z_{P_{max}}$	Y_{var}	$Y_{P_{max}}$	X_{var}	$X_{N_{max}}$	X_{ApEn}	X_{cross}	Z_{var}
1		+	+	+	+	+	+								
2		+	+	+	+	+	+	+							
3		+	+	+	+	+	+	+	+	+					
4		+	+	+	+	+	+	+	+	+	+	+			
5		+	+	+	+	+	+	+	+	+	+	+	+	+	
6		+	+	+	+	+	+	+	+	+	+	+	+	+	+
7		+	+	+	+	+	+	+	+	+			+	+	
8		+	+	+	+	+	+	+	+	+			+	+	+

TABLE III
VEHICLE CLASSIFICATION RESULTS UNDER DIFFERENT CONDITIONS

Condition	Feature evaluator	Run-time ^[1] (s)	Number of features	Classifier	Acc (%)	Clopper-Pearson interval ^[2]
Feature extraction	FPE-filter	4.49	11	C4.5	89.75	[0.8733 , 0.9186]
		4.04	14	KNN	95.08	[0.9326 , 0.9653]
		3.94	11	SVM	95.36	[0.9357 , 0.9676]
		4.01	11	RF	94.67	[0.9279 , 0.9618]
	None	0	34	C4.5	91.39	[0.8912 , 0.9332]
		0	34	KNN	93.99	[0.9201 , 0.9560]
		0	34	SVM	95.22	[0.9341 , 0.9665]
		0	34	RF	95.22	[0.9341 , 0.9665]
No feature extraction	FPE-filter	35.26	28	C4.5	82.76	[0.7983 , 0.8543]
		35.26	28	KNN	84.82	[0.8201 , 0.8734]
		35.78	35	SVM	90.29	[0.8791 , 0.9234]
		36.01	36	RF	88.78	[0.8627 , 0.9098]
	None	0	846	C4.5	80.85	[0.7780 , 0.8364]
		0	846	KNN	82.63	[0.7968 , 0.8531]
		0	846	SVM	88.51	[0.8597 , 0.9073]
		0	846	RF	88.37	[0.8582 , 0.9061]

^[1] Time spent on feature selection.

^[2] Significance level $\alpha = 0.05$.

feature dimensions of the entire data set the same. Therefore, each data point in a signal represents a feature of the data set. In this paper, the number of features of the data set under the no feature extraction condition is 846.

As can be seen in Table III, under the condition of using data processed by feature extraction, feature selection slightly improves recognition performance of the KNN and SVM classifiers but slightly decreases performance of the C4.5 and RF classifiers. From the perspective of run-time efficiency and the number of selected features, FPE-Filter selects less than half of the total number of features in about 4 seconds, but achieves a classification accuracy that is not inferior to no-feature-selection. Under the condition of no feature extraction, the accuracies of the four classifiers based on FPE-Filter are improved compared with no-feature-selection, and the number of features used has been reduced from more than 840 to only about 30. In addition, under the same conditions, the data based on feature extraction performs better on the classifier than the data without feature extraction, both in terms of classification accuracy and the number of features used. In summary,

the experimental results in Table III verify the high efficiency of FPE-Filter and the necessity of using feature extraction. Furthermore, this paper uses Clopper-Pearson interval to assess the confidence range of each classification accuracy. It can be seen that the confidence intervals are kept in a small range. And according to the average and the best value of the classification accuracy, we can also conclude that SVM is more suitable for vehicle classification than RF, KNN, and C4.5. Table IV lists the classification results for the three types of vehicles that correspond to the SVM's best classification accuracy. The corresponding feature subset is as follows

$$F_{best} = [X_{max}, X_{P_{max}}, X_{P_{min}}, X_{N_{max}}, X_{var}, Y_{max}, Y_{P_{max}}, Y_{P_{min}}, Y_{var}, Z_{P_{max}}, Z_{P_{min}}] \quad (8)$$

As can be seen from Table IV, the classification accuracy of the three types of vehicles has reached more than 90%, of which SAS achieves nearly 100%. The main misclassified samples are mainly concentrated between TRUCK and BUS. During the experiment, we found that if the distance between the SUV and the sensor node is very close, the magnetic field

TABLE IV
THE BEST CLASSIFICATION RESULT OF SVM

CLASS	Result				
	TRUCK	SAS	BUS	Total	Accuracy
TRUCK	196	0	17	213	92.0%
SAS	0	295	2	297	99.3%
BUS	13	2	207	222	93.2%
Total	209	297	226	732	95.4%

disturbance caused by the SUV may reach the level of the BUS. But the experimental results show that this influencing factor can be offset by some features. In general, the high classification accuracy of the classification model verifies the effectiveness of the FPE-Filter algorithm.

VII. CONCLUSION

This paper studies the effectiveness of feature selection for the vehicle classification based on magnetic sensors from the perspective of data mining. A novel and concise feature selection algorithm, FPE-Filter, is used to select the optimal subset from the 34 extracted features. The feature selection algorithm needs to preset two input parameters, namely a reliefF threshold W_t and a Pearson correlation coefficient threshold R_t . We compare the classification performance of the classifiers in the condition of feature extraction and no feature extraction, respectively. The experimental results show that the feature extraction can significantly improve the accuracy of the classifiers. Under the premise of using feature extraction, we also compare the classification performance of the four classifiers in the condition of FPE-filter and no feature selection, respectively. The experimental results show that FPE-filter allows using only approximately one third of all considered features without decreasing classification accuracy. In the four types of classifiers that participated in the comparison, SVM showed the best performance in both the average accuracy and the highest accuracy. Although the classifier based on FPE-Filter is not significantly better than the classifier without feature selection, it still has practical application significance due to its fewer feature numbers and high classification accuracy. If more vehicle data is collected, we can get more accurate and reliable classification results. The experimental results reveal the preference of different classifiers for different features, so finding efficient features is also one of the keys to improve classification performance. Under the experimental conditions of this paper, we did not consider the interference factors of adjacent lanes. Congested traffic conditions can also make vehicle classification more difficult. These will be the focus of our future research.

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